LECTURE ACTIVITY 19: CONSERVING HOTEL TOWELS, PART II

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from "Introduction to Statistical Investigations," by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

1) Identify the value of the chi-square test statistic using the Multiple Proportions applet.
2) Recognize that larger values of the statistic mean more evidence against the null hypothesis and why the distribution of the chi-square statistic is non-negative and follows a right-skewed distribution.
3) Conduct a simulation-based analysis using the chi-square statistic when evaluating claims about multiple proportions.
4) Conduct a theory-based analysis when evaluating claims about multiple proportions.
5) Identify whether the validity conditions are met for a chi-square test.
6) State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the p-value and study design.
7) Conduct a follow-up analysis by calculating confidence intervals for all pairwise comparisons between the groups.

In the previous activity "CONSERVING HOTEL TOWELS" we compared multiple proportions by using the Mean Group Diff statistic and following a simulation-based approach. We will go back to the same context but now focusing on a different statistic (chi-square) and using theory-based approach.

Step 1: Formulate a Research Question

Same research question as before "Do different socially normed signs produce different rates of towel reuse?".

Step 2: Design a Study and Collect Data

As a reminder, the study was a randomized experiment to investigate how different phrasings on signs placed on bathroom towel racks impacted guests' towel reuse behavior. In particular, the researchers were interested in evaluating how messages that communicated different types of social norms impacted towel reuse. One week prior to a guest staying in the room, rooms at a particular hotel were randomly assigned to receive one of five messages on a sign hung on the towel bar in the room: no social norm (none), guest identity norm, same room norm, citizens norm, and sex identity norm.

Data were collected on 1,595 instances of potential towel reuse. For each of the 1,595 instances, room attendants recorded whether the hotel guest reused their towels.

The hypotheses we are investigating are

$$\begin{cases} H_0: & \pi_{\text{none}} = \pi_{\text{guest}} = \pi_{\text{room}} = \pi_{\text{citizens}} = \pi_{\text{sex}} \\ H_A: & \text{At least one population proportion of towel-reuse instances differs across the message types} \end{cases}$$

Step 3: Explore the Data

As a reminder, here are the sample proportions as well as a segmented bar graph:

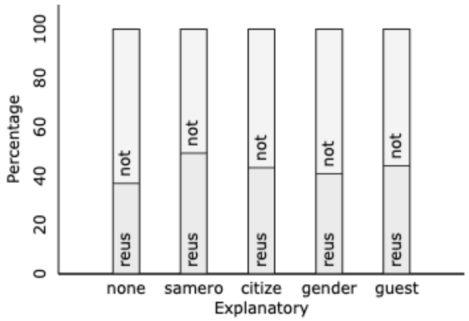


Figure 1: Segmented Bargraph

Message	Proportion	Total
None	0.370	305
Same Room	0.493	306
Citizen	0.434	334
Sex	0.410	310
Guest	0.441	340

Table 1: Data Summary

Step 4: Draw Inferences Beyond the Data

In the last activity we used the Mean Group Diff statistic. This statistic is fairly easy to understand but is not widely used in part because it cannot be easily predicted theoretically. A more commonly used statistic is called a **chi-square** (χ^2; pronounced "kai-square") statistic. You can think of this statistic as measuring how far the group proportions of success are from the overall proportion of success. These deviations from the overall proportion are then squared and summed together across all of the explanatory variable groups.

$$\chi^2 = \sum_{i=1}^{n} \left[\frac{\hat{p}_i - \hat{p}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_i}}} \right]^2 \tag{1}$$

Equation (1) can seem a bit scary at first, so let's break it down.

1) First calculate $\hat{p}$, defined to be the overall proportion of successes

2) Then, calculate $\sqrt{\hat{p}(1-\hat{p})/n_i}$ for every group i, where n_i denotes the sample size of group i.

3) Then, standardize each sample proportion $\hat{p}_i$ (the sample proportion of successes in group i) using the standard deviation from the previous step

4) Square all the values from the previous step and sum them together.

	None	Same Room	Citizen	Sex	Guest
Sample Size (n_i)	$n_1 =$ _____	$n_2 =$ _____	$n_3 =$ _____	$n_4 =$ _____	$n_5 =$ _____
Conditional Proportion	$\hat{p}_1 =$ _____	$\hat{p}_2 =$ _____	$\hat{p}_3 =$ _____	$\hat{p}_4 =$ _____	$\hat{p}_5 =$ _____
Standard Deviation					
Standardized Statistic					
χ^2 Statistic:					

Table 2: Calculating the χ^2 statistic

Action Item 1

Fill in the first two rows of Table (2) using the sample data provided in Table (1) above.

Action Item 2

Calculate the overall proportion $\hat{p}$. (Hint: in this case, this should be the proportion of all rooms that reused towels.)

Action Item 3

Calculate the standard deviations of the group proportions. For example, for Group 1:

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{305}} \approx 0.028$$

Action Item 4

Calculate the standardized z-statistics within each group. That is, take each of the five group proportions, subtract the overall proportion ($\hat{p}$) and then divide the result by the standard deviation you found for that group. For example, for Group 1:

$$\frac{\hat{p}_1 - \hat{p}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1}}} = \frac{0.370 - \hat{p}}{0.028} \approx -2.10$$

Action Item 5

Calculate the value χ^2 statistic. You should get around 10.15.

Question 1

(a) As mentioned above, this statistic χ^2 can be considered as measuring **how far the group proportions of success are from the overall proportion of success**. With this idea in mind, what kinds of values (larger, small, positive, negative) will the chi-square statistic have if the null hypothesis is true? What kinds of values will provide evidence against the null hypothesis?

(b) Based on the calculation of the χ^2 statistic, will this statistic ever be negative? Explain.

(c) Navigate back to the Multiple Proportions applet, and select the "**Towel**" dataset from the dropdown menu. Click "**Use Table**." Then, under the "**Statistic**" menu, select χ^2. What does the applet report for the observed chi-square statistic?

Pro Tip

We will very rarely ask you to calculate the chi-square statistic by hand, and opt instead to let the applet calculate it for us :)

While we're in the applet, let's get a quick feel for simulation-based inference using the chi-square statistic.

Question 2

Before we conduct the simulation, what do you think will be the shape of the distribution of many χ^2 statistic values, under the null hypothesis?

1) Make sure the **Statistic** option is set to $\boldsymbol{\chi^2}$
2) Check the box next to **Show Shuffle Options**.
3) Press **Shuffle** to perform one shuffle of the response variable values.
4) Change the **Number of Shuffles** from 1 to 9,999.
5) Press **Shuffle**

Question 3

The applet will also now display the null distribution for the simulated χ^2 statistics rather than the simulated MEAN GROUP DIFF statistics. Determine the *p*-value based observed χ^2 statistic. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) *p*-value.

- My approximate simulation-based *p*-value is: _____

Here is my screenshot:

As a reminder, here is the output we got when we used the mean group difference statistic (in the last lecture activity):

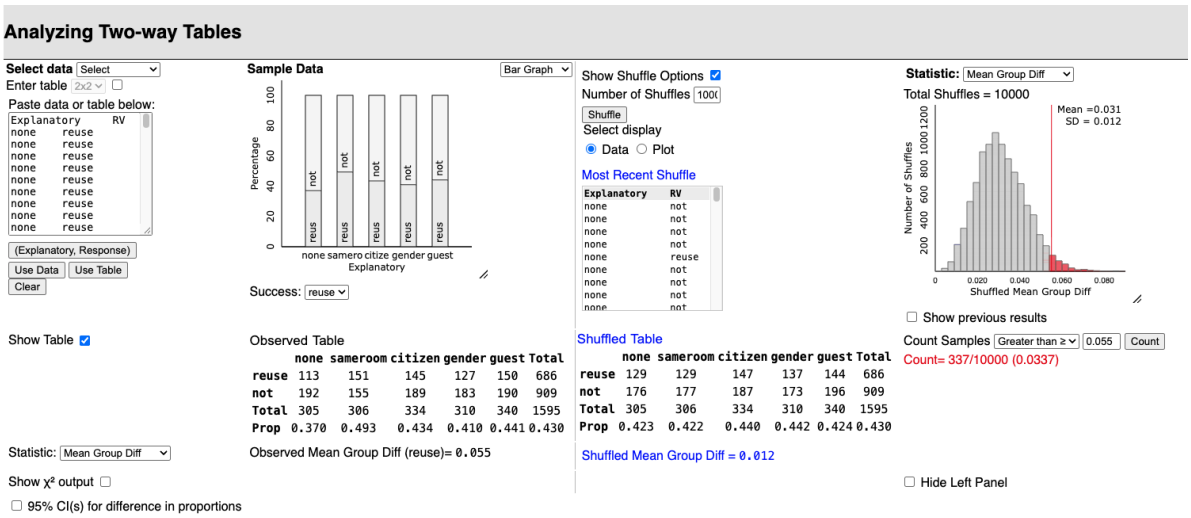


Figure 2: Output from the Multiple Proportions applet, using the Mean Group Diff statistic

Question 4

How does the *p*-value based on the observed χ^2 statistic compare to the one based on the observed mean group difference statistic? Are they similar? Is your conclusion about strength of evidence provided by the data against the null hypothesis similar?

Theory-Based Approach

One of the biggest advantages of using the χ^2 statistic (as opposed to using the mean square difference statistic) is that its null distribution can be predicted by a theoretical distribution. So, once again a theory-based approach could be used without conducting a simulation in the first place.

Question 5

Based on what we saw in our simulation, do you think the null distribution of the χ^2 statistic will be normal? Explain briefly.

Indeed, the null distribution of the χ^2 statistic follows something known as the **χ^2 distribution**. Whereas the normal distribution is symmetric and bell-shaped, the χ^2 distribution is asymmetric and right-skewed.

Return to the applet, and Below the graph of the simulated chi-square statistics, check the box to **Overlay Chisquare distribution**. You should see that the theoretical distribution matches the distribution of simulated statistics reasonably well, and should also see a theory-based *p*-value appear in orange near the bottom.

Question 6

How does the theory-based p-value compare to what you found in Question 3 above? Does it lead you to a similar conclusion about strength of evidence?

Concept Overview

Validity Conditions for the Theory-Based Approach. Like all theory-based tests, this one also comes with the condition of having a large sample size. We will (very conservatively) consider a sample size large if the sample data include at least 10 observations in each interior cell of the two-way table.

Question 7

Go back to the applet and look at the **Observed Table** on the left part of the applet. You should notice that there are 10 middle cells in the table. Are all of these cells at least 10? In other words, is the validity condition for a theory-based chi-square test satisfied for these data? Explain.

Step 5: Formulate Conclusions

At this point, you should have noticed that regardless of whether we're using a simulation-based approach with the Mean Group Diff statistic or a a theory-based approach with the χ^2 statistic, we obtain the same conclusion: at least one population proportion of guests reusing towel (π) is different across message types.

Question 8

Think as if you were the manager of the hotel, what other information would you want to know?

A sensible next step is to try to identify which population proportion differs significantly from which others. We can do this by producing confidence intervals for pairwise differences in population proportions.

- Go back to the applet and check the box next to **Compute 95% CI(s) for difference in proportions** (bottom left side of the screen).

Question 9

A total of 10 intervals were produced. Why do we have 10 intervals instead of only 1?

Pro Tip

When comparing k groups, the total number of pairwise comparisons that can be made is

$$\frac{k \cdot (k-1)}{2}$$

For example, how many pairwise comparisons can be made among 5 groups?

Question 10

For the interval comparing NONE and SAME ROOM messages, write a one-sentence interpretation of the interval being very clear what is supposed to be captured inside the interval and which condition has a more positive effect on towel use and by how much.

Question 11

For the interval comparing NONE and SAME ROOM messages, write a one-sentence interpretation of the interval being very clear what is supposed to be captured inside the interval and which condition has a more positive effect on towel use and by how much.

Question 12

Write a report including (1) a recommendation for the hotel manager about whether it matters what message is used, (2) which message should be used, and (3) how large of an impact the best message would be expected to have. Try writing your report using simpler language that the manager, who might not have knowledge of statistics, would understand. **Tip**: pay attention to the possible size of the differences in population proportions you found.

Part 2: Physical Activity and Aging

An eight-year cohort study of 3,454 initially disease-free participants (ages around 60 years or more) recorded the extent of physical activity the participants engaged in and whether or not they were showing signs of "healthy aging" (British Journal of Sports Medicine, 2013). The research question is: Is there a relationship between healthy aging and activity level?

Here's the two-way table of the data. You can also access a copy-paste-able version by clicking here.

	Activity Level			
	Inactive	**Moderate**	**Vigorous**	**Total**
Healthy Aging ("success")	55	345	265	665
Not Healthy Aging ("failure")	598	1347	844	2789
Total	653	1692	1109	3454

Table 3: Two-Way Table of Data

The null and alternative hypothesis (in terms of associations) to be tested in this study are:

- H_0: There is no association between activity level and healthy aging, in the population.
- H_A: There is an association between activity level and healthy aging, in the population.

Question 13

(a) Are the validity conditions met to conduct a theory-based test to answer the research question? Do not just state the validity conditions! Use the data from the sample to explain how the validity conditions are or are not met.

(b) What is the name of the statistic used in the theory-based approach when comparing multiple proportions?

Now we will add the data to the Multiple Proportions applet.

1) Open this data file (linked above) and copy it (Command/Control + A to "select all," and Command/Control + C to "copy")
2) Go to the Multiple Proportions applet.
3) Press **Clear** to erase the default data.
4) Paste the copied data into the "**Data**" window (Command/Control + V to "paste")
5) Press Use **Data**
6) Choose the appropriate statistic com the pull-down menu (bottom left).

Question 14

Determine the theory-based *p*-value. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) *p*-value

- The value of the statistic is: _____
- My theory-based *p*-value is: _____

Here is my screenshot:

Question 15

Based on your analysis, give an overall conclusion in context. Make sure you refer to the parameter of interest in your conclusion.

Question 16

Generalization: Comment on how the sample was obtained and to which population your conclusion can be generalized.

Question 17

Causation: Comment on how the study was designed and whether a cause-and-effect conclusion is warranted from this study

Question 18

If appropriate, conduct the follow-up analysis, report the confidence intervals, and report which activity levels differ significantly.

If not appropriate, explain why it is not particularly useful to look at confidence intervals for the parameter(s) of interest in the context of this study

Question 19

Based on the results of this study, is it correct to say that each level of additional activity is associated with significantly more healthy aging? Explain using your answer to the previous question.